



Clean Our Sea

Basics of Mechatronics MCT 231

Team Members

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1. Introduction

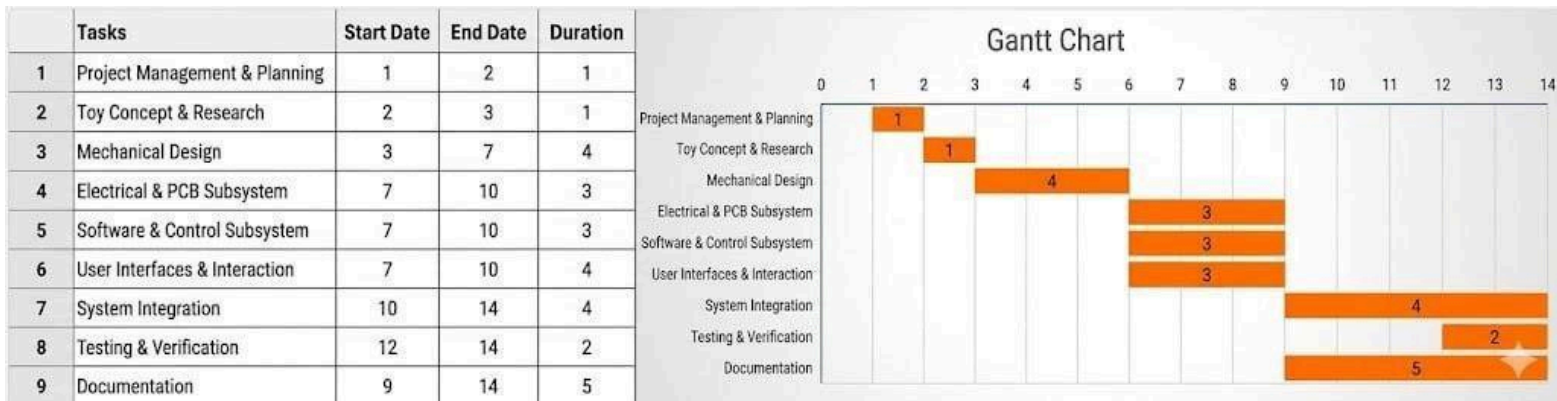
In recent years, environmental awareness has become a critical component of modern education, particularly for younger generations who will inherit the challenges of a changing world. One of the most pressing issues facing our planet is marine pollution, driven largely by improper waste disposal and a lack of public awareness about its consequences on aquatic ecosystems.

This project, developed as part of the Basics of Mechatronics course at Ain Shams University, presents an interactive educational game titled "Clean Our Sea" — designed to engage children in learning about environmental responsibility through play. The system integrates core mechatronics principles, including color-based object sorting, an automated lifting mechanism, and a shooting actuator, to simulate a real-world scenario where a child must classify and dispose of waste correctly to protect the ocean.

The game operates on a meaningful premise: colored balls represent different categories of objects found in marine environments. Red balls symbolize fish — living marine life that must be preserved — while green balls represent garbage that pollutes the sea. The child is challenged to identify the waste and shoot it into a designated trash bin, reinforcing the message that individual actions directly impact ocean health.

By combining physical interaction with an educational message, this project demonstrates how mechatronic systems can serve not only industrial and engineering purposes but also social and pedagogical goals. The integration of sensors, actuators, and control logic in a game format makes abstract environmental concepts tangible and accessible to young audiences.

2. Planning



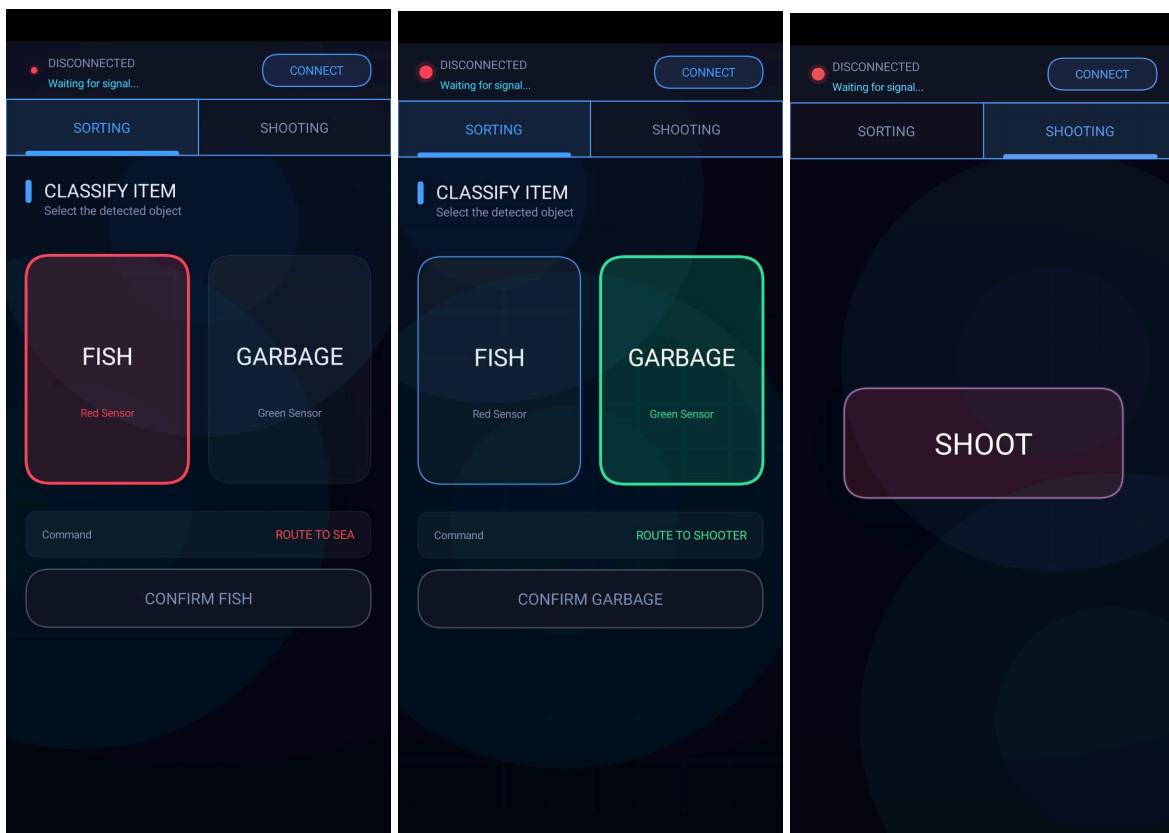
3. Theme

3.1 Toy Theme





3.2 User Interface



4. User Research

The target audience for this toy is children between the ages of 2 and 10 years old. To design an effective and engaging product, the team conducted research into the developmental needs and preferences of this age group, focusing on the following areas:

4.1 Cognitive and Motor Development

Children in the 2–10 age range are in active stages of cognitive and physical development. Key characteristics considered include:

- Ages 2–4: Attracted to bright colors, simple cause-and-effect interactions, and tactile experiences.
- Ages 5–7: Capable of understanding simple rules, developing fine motor skills, and beginning to grasp cause-and-consequence concepts.
- Ages 8–10: Able to engage with more complex gameplay, problem-solving, and absorb educational content through interaction.

4.2 Environmental Awareness in Early Education

Research shows that environmental habits and values are most effectively instilled during early childhood. Gamified learning tools have proven to increase retention of environmental concepts in children significantly better than passive instruction. Our toy leverages this by embedding the message — "garbage does not belong in the ocean" — directly into the gameplay loop.

4.3 Design Insights from Research

Based on our research, the following design principles were adopted:

- Use of distinct colors (red and green) to make classification intuitive even for very young children.
- Physical interaction (shooting mechanism) to engage motor skills and make the experience memorable.
- Clear visual feedback — a successful shot results in the ball landing in the trash bin, reinforcing the correct behavior.
- Safe materials and rounded edges to comply with toy safety standards for young children.

5. Design Challenge

The primary challenge addressed by this project is the lack of engaging, hands-on tools for teaching children about environmental responsibility — specifically marine pollution caused by improper waste disposal.

5.1 Problem Statement

Ocean pollution is one of the most critical environmental challenges of our time. Millions of tons of garbage enter the world's oceans annually, threatening marine biodiversity and ecosystems. Despite widespread awareness campaigns, behavioral change — especially in habits around waste disposal — remains slow. Instilling responsible environmental behavior from an early age is therefore essential.

5.2 Why This Problem Matters

- Marine pollution directly threatens fish populations and entire aquatic ecosystems.
- Children who develop environmental awareness early are more likely to carry responsible habits into adulthood.
- Traditional educational methods (books, videos) are passive and less effective for early childhood learning compared to interactive, play-based approaches.

5.3 Engineering Challenge

From a mechatronics perspective, the design challenge was to build a reliable, child-safe, and functional system that could:

- Accurately detect and sort colored objects in real time using a sensor.
- Physically lift and position objects without jamming or requiring manual reset.
- Execute a repeatable, controlled shooting action that is safe for children.
- Integrate all subsystems (sensing, lifting, shooting) into a seamless, responsive game experience.

6. Ideation

The ideation process began with brainstorming sessions focused on two core questions: How can we make environmental education fun for children? And how can we demonstrate mechatronics principles in a meaningful, real-world context?

6.1 Concept Generation

Multiple concepts were initially proposed, including a recycling sorting machine, a marine clean-up simulation, and a factory conveyor game. The team converged on the ocean/garbage concept because it combined a clear, relatable message with all three required mechanical subsystems: sorting, lifting, and shooting.

6.2 Final Concept Selection

The final concept was selected based on three criteria: feasibility within the project timeline, alignment with the educational goal, and ability to demonstrate all three required mechatronic subsystems. The "Clean Our Sea" concept scored highest across all criteria and was unanimously selected by the team.

7. Toy Description and Functionality

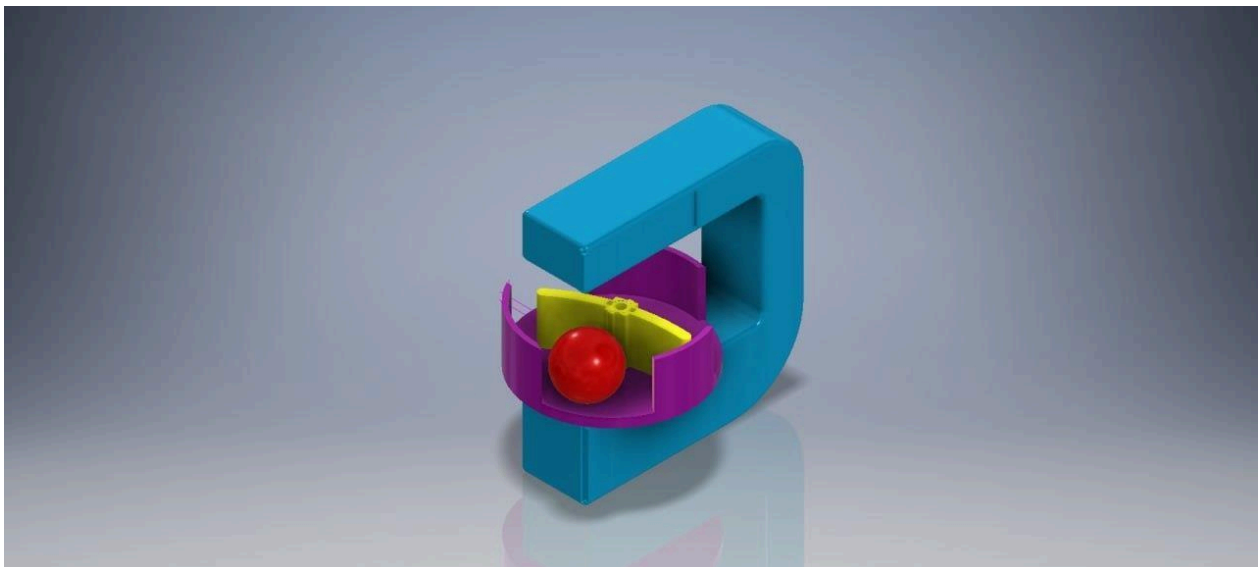
The "Clean Our Sea" toy is a table-top mechatronic game consisting of three integrated subsystems: a sorting mechanism, a lifting mechanism, and a shooting mechanism. Together, these systems simulate the process of identifying and removing waste from a marine environment.

7.1 Mechanical Design

The mechanical design was developed using Autodesk Inventor. The toy consists of the following mechanical assemblies:

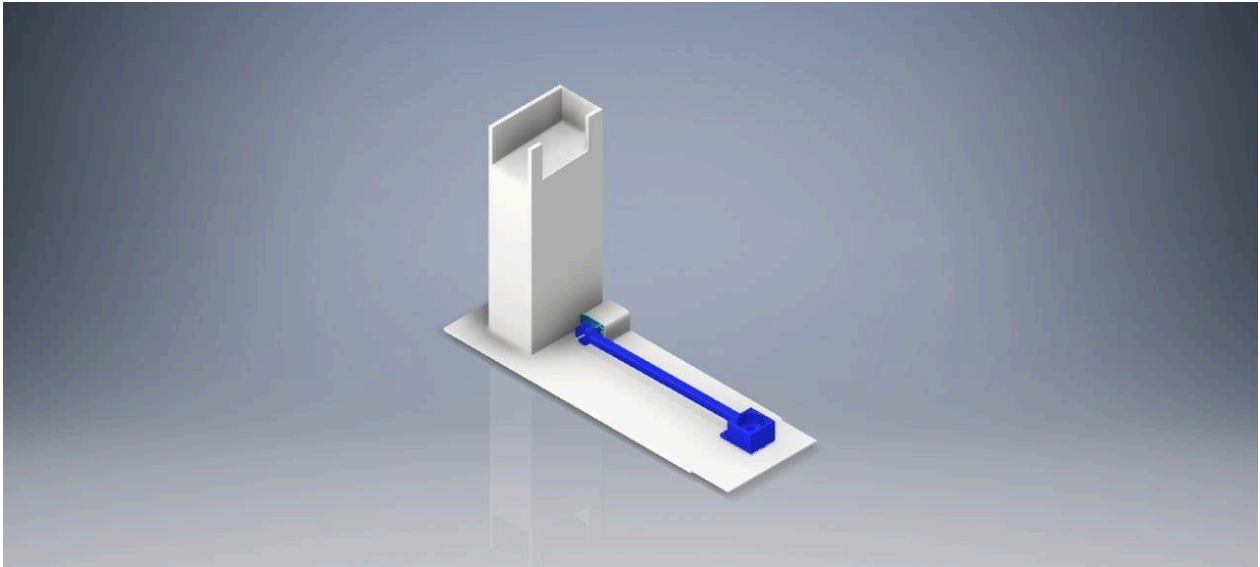
7.1.2 Sorting Mechanism:

Colored balls are fed into the system via a chute or conveyor belt. A color sensor positioned at the entry point reads the ball's color and signals the control system. Based on the signal, a gate or deflector routes the ball either to the lifting arm (green = garbage) or to a bypass path (red = fish).



7.1.2 Lifting Mechanism:

Once a ball is identified, a motorized lifting arm elevates the ball to the shooting position. The arm is driven by a Servo motor to ensure consistent and repeatable lifting to the correct height.

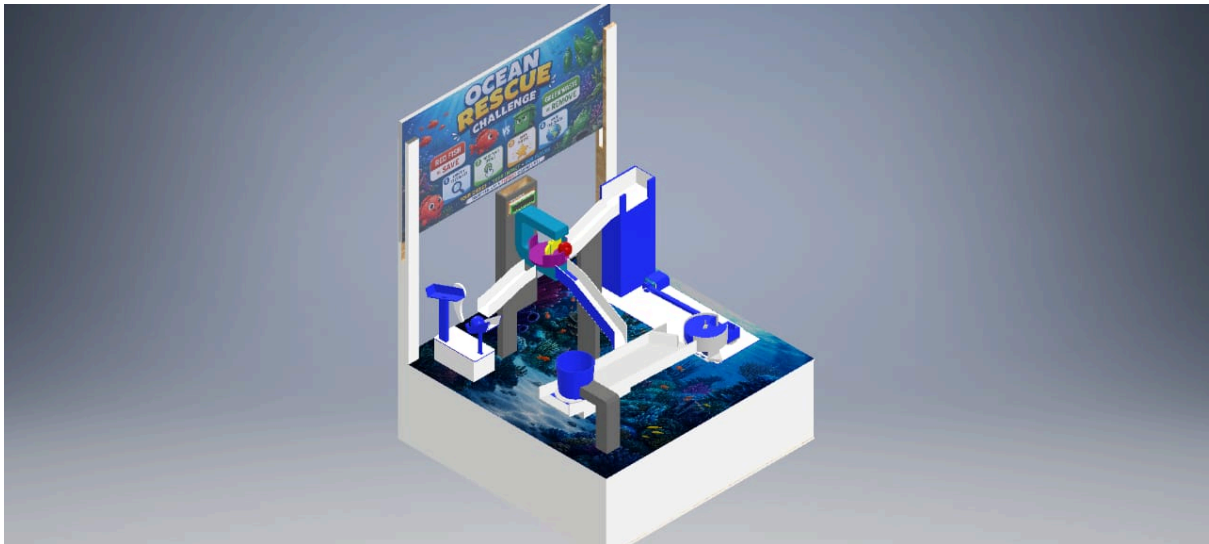


7.1.3 Shooting Mechanism:

The shooting mechanism is actuated by a DC motor. The child can trigger the shoot via a push button, making the experience interactive. The trajectory is calibrated to land the ball in the trash bin placed at a fixed distance.



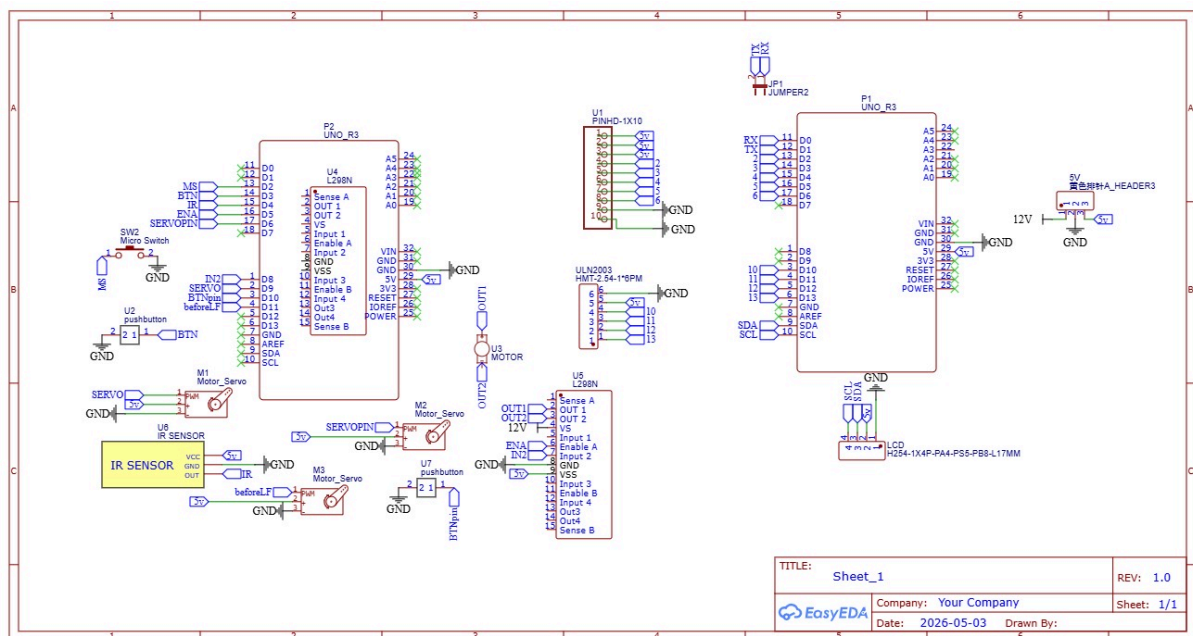
7.1.4 Fully Integrated Toy



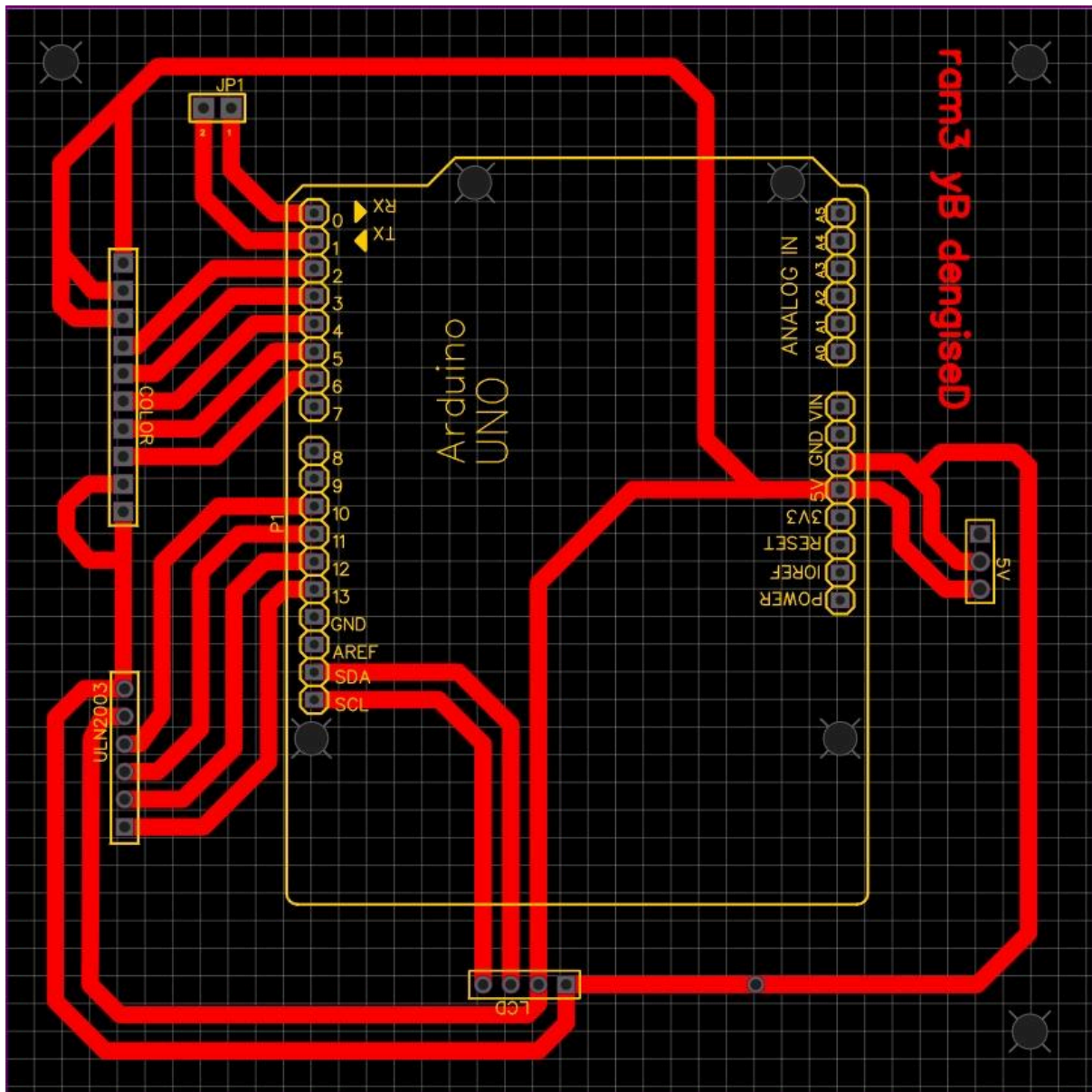
7.2 Electrical Design

The electrical design was implemented using Arduino UNO as the main microcontroller and was simulated/designed using Proteus.

7.2.1 Schematic for toy



7.2.2 Pcb Design



7.3 Sensors & Actuators System

7.3.1 Sorting Subsystem

Color Sensor (TCS3200)

- Positioned at the ball entry point
- Emits white light and reads the reflected wavelength
- Classifies ball as **Red (Fish)** → fish path, or **Green (Garbage)** → lifting zone
- Output frequency is processed by Arduino for color decision

Stepper Motor

- Receives classification signal from Arduino
- Precisely rotates a directional gate/platform to move the ball to its correct path
- Used for its **open-loop positional accuracy** — no extra feedback sensor needed
- Ensures ball lands exactly in the right zone without overshoot

7.3.2 Lifting Subsystem

IR Sensor

- Mounted at the base of the lifting zone
- Detects ball arrival before lifting begins
- Acts as a **trigger** — servo motors do not activate until IR confirms ball is in position
- Prevents false actuations and mechanical errors

Servo Motor(s)

- Activated upon IR sensor confirmation
- Lifts the ball upward to the shooting position
- Used for **controlled, repeatable angular movement** with defined start/end positions
- Multiple servos can be used at different arm joints depending on mechanical design

7.3.3 Shooting Subsystem

DC Motor

- Connected to a swinging/rotating arm
- Triggered by child button press after lifting is complete
- Arm sweeps and **strikes the ball**, launching it toward the trash bin
- Chosen for its **high-speed rotational output** delivering the required impact force

7.3.4 Feedback Subsystem

Buzzer

- Provides **audio feedback** to the child during gameplay
- Triggers on: successful shot into bin, wrong ball detected, system ready state
- Reinforces correct behavior (shooting garbage) with positive sound cues
- Connected directly to Arduino digital output pin

LCD Display + I2C Module

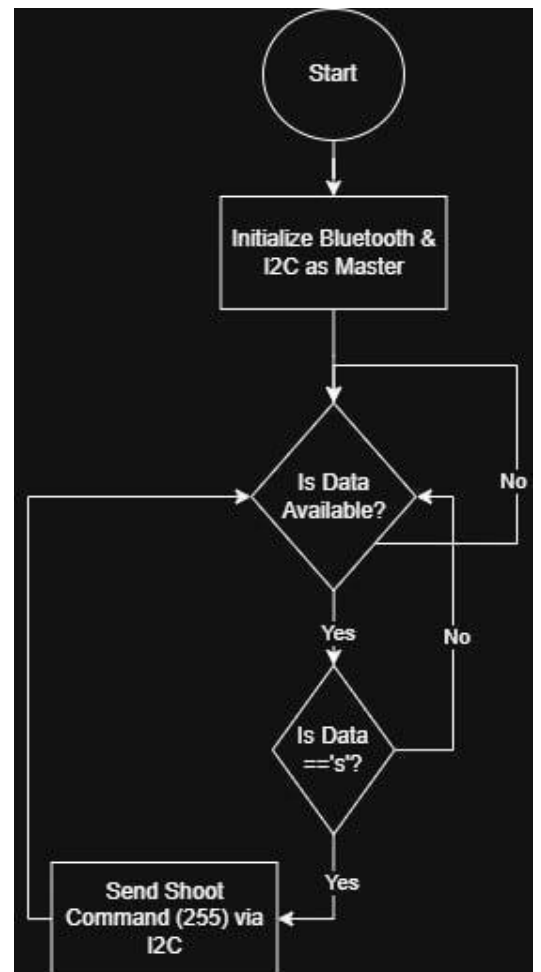
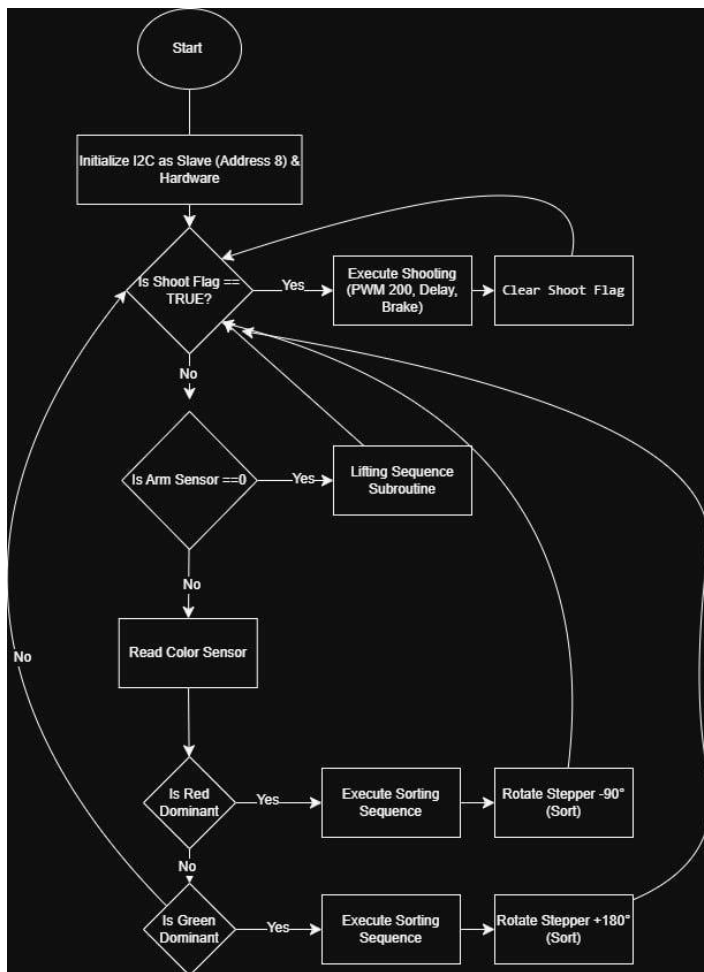
- Displays real-time **score, instructions, and system status** to the player
- I2C interface reduces wiring complexity — only 2 signal wires to Arduino (SDA, SCL)
- Shows messages such as: *"Shoot the garbage!"*, *"Great job!"*, current score count
- Useful for guiding younger children through the gameplay steps

7.3.5 Power Management Subsystem

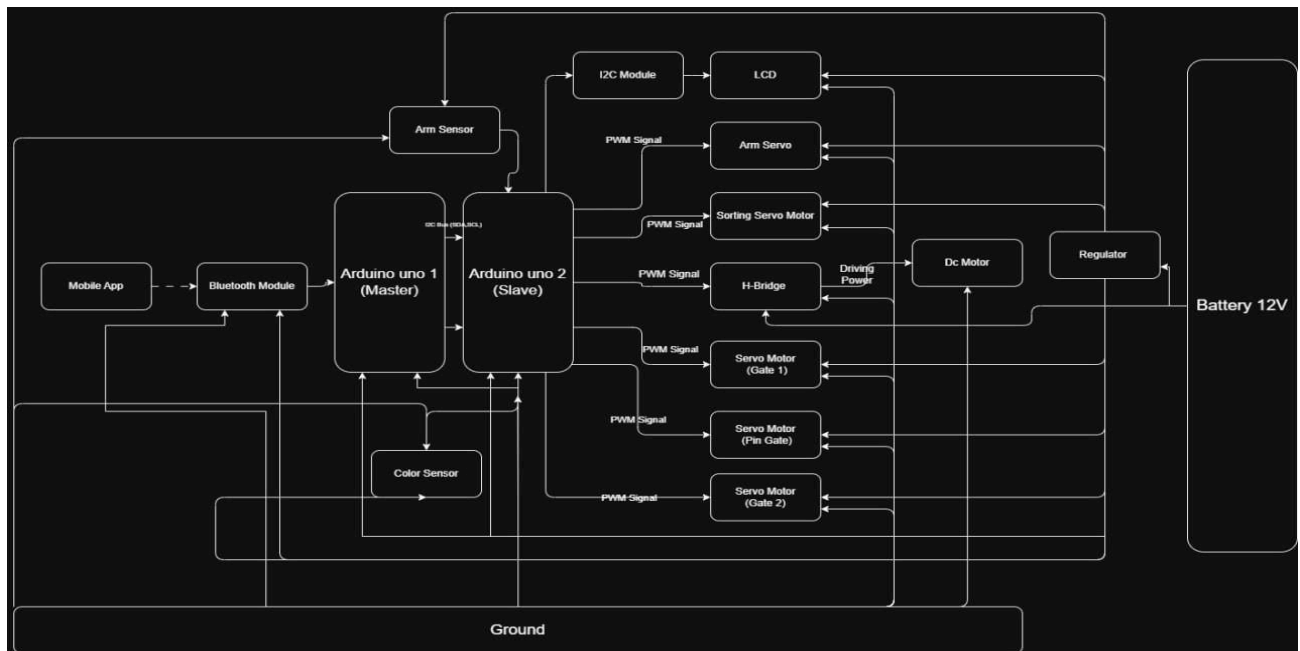
Voltage Regulation

- System operates on a **12V main supply**
- Step-down regulators provide **5V** for Arduino, sensors, LCD, and buzzer
- **Separate power rails** for motors (12V) and logic components (5V) to prevent voltage noise interference
- Protects sensitive components from motor-induced voltage spikes

8. Flow charts



9. Block Diagram:



10. Safety Considerations

Child safety is a primary design requirement. The following safety measures were implemented in accordance with standard toy safety guidelines:

- All exposed edges are rounded or covered with soft material to prevent cuts or abrasions.
- Ball diameter (4cm) exceeds the choking hazard threshold for children under 3 years old.
- The shooting mechanism is calibrated to a low-force launch that cannot cause injury.
- All electrical components are enclosed and inaccessible to the child during play.
- Operating voltage is limited to 5V at the child-accessible interface; high-voltage components are fully enclosed.
- Non-toxic materials were used for all surfaces that children may touch.
- The system includes an auto-stop feature if a ball jams to prevent motor burnout or mechanical damage.

11. Educational Impact

"Clean Our Sea" is designed not only as an entertaining toy but as a learning tool with measurable educational outcomes:

11.1 Environmental Awareness

The core gameplay loop — identify garbage, shoot it in the bin, protect the fish — directly teaches children that waste does not belong in the ocean and that individual actions have real consequences for marine life. This message is reinforced through every successful play cycle.

11.2 Cognitive Skills

- Color recognition and classification.
- Cause-and-effect reasoning (if I shoot the garbage, the ocean stays clean).
- Decision-making under time pressure (which ball is coming next?).

11.3 Motor Skills

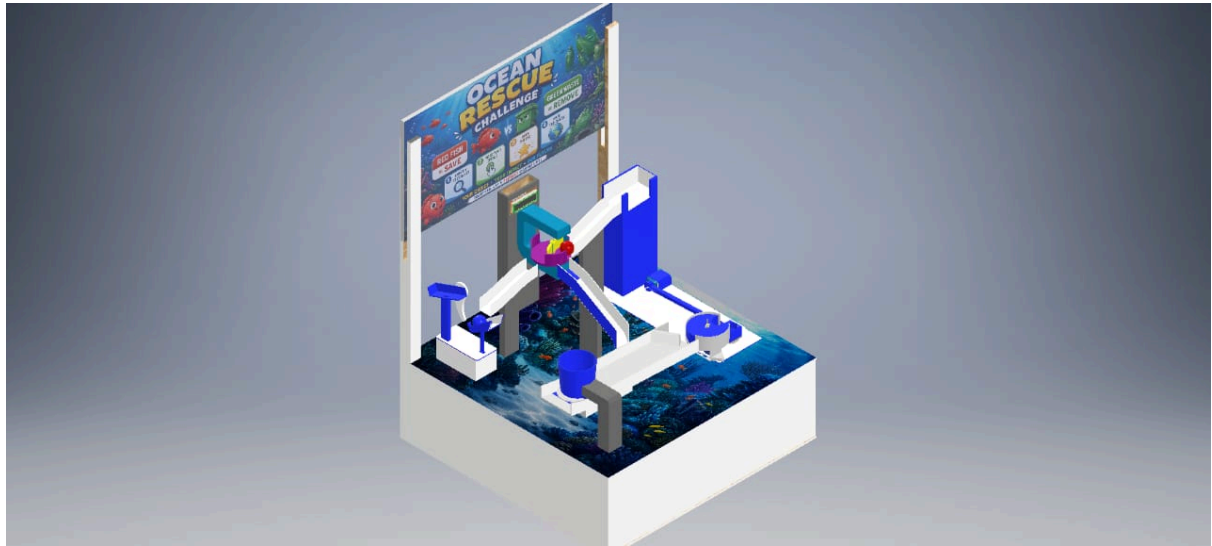
- Hand-eye coordination via the shooting interaction.
- Fine motor development through button pressing and game interaction.

11.4 STEM Foundation

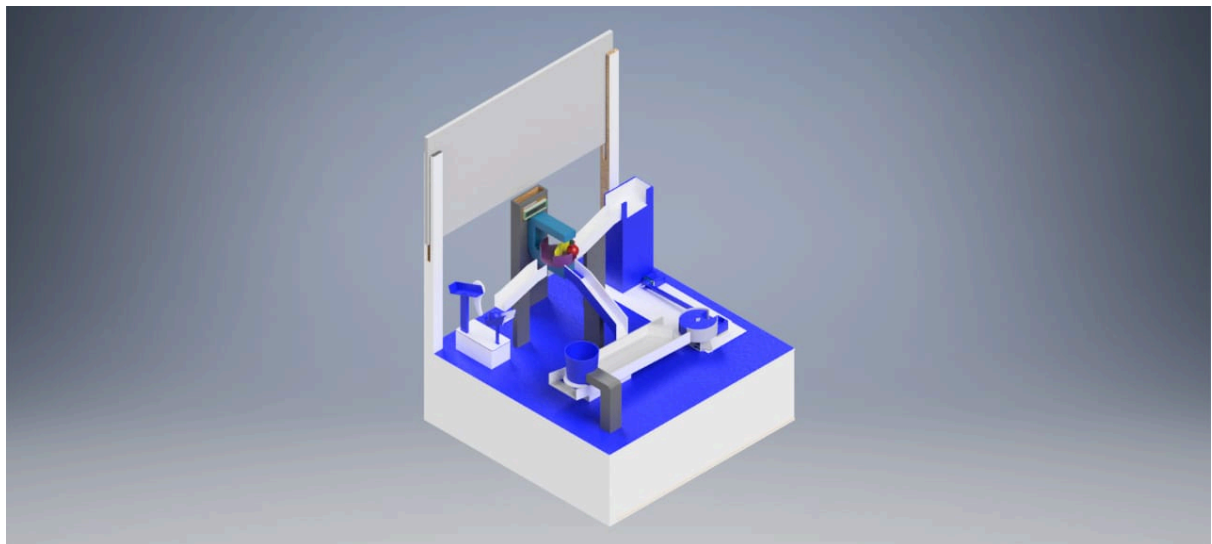
For older children in the 7–10 range, the visible mechanical and electronic components of the toy serve as an introduction to basic engineering concepts, sparking curiosity about how machines work and laying the groundwork for future STEM interest.

12. Final Product

The final version of "Clean Our Sea" represents the culmination of the design, prototyping, and iteration process described in this report. The system operates reliably across all three subsystems and delivers a consistent, engaging play experience.



Render Photo



13. Conclusion

This project successfully demonstrates the integration of mechatronics principles — sensing, actuation, and control — in a child-friendly, educationally meaningful product. "Clean Our

Sea" achieves its dual goals of delivering an engaging play experience and communicating a powerful environmental message to young users.

The project highlights how engineering disciplines can be applied beyond industrial contexts to address social challenges, particularly in early childhood education. The sorting, lifting, and shooting subsystems work in harmony to create a seamless and reliable game experience that is both safe and stimulating for children aged 2–10.

Future development could include adding a scoring display, sound effects for correct/incorrect actions, and expanding the color categories to teach broader waste classification concepts (e.g., plastic, glass, organic waste). Integration with a mobile app could also allow parents and teachers to track a child's progress over time.

The team is confident that "Clean Our Sea" has the potential to contribute meaningfully to environmental education and serves as a strong example of purposeful mechatronics design.

14.

<https://github.com/youssefmakhlouf777/Basics-Project>

Thank you